

Photosynthesis Float-Off Playoffs

Make leaf disks float by making oxygen

At a glance: Make leaf disks float by producing oxygen. Plan about 60 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: photosynthesis, oxygen production, variables, rate (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- Spinach leaves: 10 disks per team (about 1 bag/bunch for the camp; use fresh leaves)
- Baking soda solution: 0.1% (0.5 g baking soda per 500 mL water) plus a few drops of dish soap; about 160 mL per team per run, mix about 1.5 to 2 L total
- Liquid dish soap: a few drops per 500 mL batch as a wetting agent so the disks sink; the assay fails without it (MISSING from the buy list; add 1 small bottle)
- Cups: 1 clear 250 mL cup per team (4 plus spares)
- Needle-free 10 mL syringes: 1 per team (4 plus 2 to 4 spares)
- Light source: 1 clamp lamp per team with a bright LED or 100 W equivalent bulb, about 12 in (30 cm) above the cup (MISSING from the buy list; borrow 4, or share the PYS-10 lights)
- Timers: 1 per team (kit/shared; or phones plus the projected timer)
- Table cover: a trash bag taped flat over each wet station to catch spills
- Hole punch or stout straws: to cut 10 uniform disks; straws work best with spinach
- Nitrile gloves: 1 pair per student (shared PPE box, already bought)

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up one station per team with the materials above, sorted and ready, including one clamp lamp per station (bright LED or 100 W equivalent) suspended about 12 in (30 cm) above the cup. Test that each lamp reaches the disks; without light the disks will not float.
- 02 Mix the 0.1% baking-soda solution (0.5 g baking soda per 500 mL water), add a few drops of dish soap and stir gently to avoid suds, cut fresh spinach disks, stage a slop bucket or compost bin, and put out the scoring sheet before testing begins.
- 03 Put out nitrile gloves and tape a trash bag over each station table before any materials are handled. No goggles are needed: a few drops of dilute baking-soda solution is not a chemical or splash hazard.
- 04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Make leaf disks float by producing oxygen." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:38

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams load 10 disks in the syringe, pull a vacuum 2 to 3 times to sink them in the soapy bicarbonate solution, pour up 150 mL in the cup under the lamp, start the timer, record floating disks each minute, and log the baseline half-float time (minutes to 5 of 10 floating) before changing anything.

Phase 6 Redesign

Each team changes exactly one variable (light distance, light brightness, or baking-soda concentration), predicts the effect, and re-runs the assay to compare the half-float time against baseline.

Phase 7 Score, debrief, cleanup

Last 12 min

Teams submit a score slip, answer a debrief question with evidence, then reset the station: pour the spent solution and spinach disks into the slop bucket or compost (or down the sink with running water), rinse the syringes and cups, wipe the tray and table dry, then remove gloves and wash hands.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: The disks will not sink, or none float after several minutes. **Fix:** If the disks will not sink, add a few more drops of dish soap and re-pull the vacuum (too little soap is the usual cause); if none float, check the lamp is on and about 12 in above the cup and that the solution has baking soda; then log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to photosynthesis makes oxygen.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why does oxygen make the disks float?
- Which variable changed the rate most?
- How is this evidence of photosynthesis?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Controlled-variable design	25
Data table	20
Explanation	10
Cleanup	10
Total	100

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES Exploratorium Photosynthetic Floatation (0.1% bicarbonate = 0.5 g per 500 mL plus a few drops of dish soap, 150 mL per cup, 10 disks, needle-free 10 mL syringe vacuum, lamp 12 in above the cup, half-float time); Brad Williamson floating leaf disk assay (0.2% recipe, hold the vacuum about 10 s, repeat 2 to 3 times).

Floating-disk data table

Print 1 per team. Load 10 disks, start the timer, and record how many are floating at the end of each minute for the baseline run, then again after your redesign. The graded data table is worth 20 points.

Team _____ The ONE variable we changed for the redesign: _____

MINUTE	BASELINE: DISKS FLOATING	REDESIGN: DISKS FLOATING
0		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Half-float time, baseline (minutes until 5 of 10 disks float): _____

Half-float time, after redesign: _____